1 I

1.1 (1)

For a single particle, there are two possible energy levels, $-\epsilon$ and ϵ. Thus, the single-particle partition function is

$$\begin{aligned} Z_1 &= e^{\beta\epsilon} + e^{-\beta\epsilon} \\ &= 2\cosh(\beta\epsilon) \\ &= 2\cosh\left(\frac{\epsilon}{k_B T}\right), \end{aligned}$$

where $\beta = 1/(k_B T)$.

Since the N particles are distinguishable and non-interacting, the total partition function is

$$\boxed{Z = Z_1^N = \left[2\cosh\left(\frac{\epsilon}{k_B T}\right)\right]^N}.$$

1.2 (2)

The internal energy can be obtained from the partition function:

$$\begin{aligned} E &= -\frac{\partial \ln Z}{\partial \beta} \\ &= k_B T^2 \frac{\partial \ln Z}{\partial T} \\ &= -N\epsilon \tanh(\beta\epsilon). \end{aligned}$$

Therefore,

$$\boxed{E = -N\epsilon \tanh\left(\frac{\epsilon}{k_B T}\right)}.$$

1.3 (3)

The heat capacity is

$$C = \frac{\partial E}{\partial T}.$$

Using the result from (2),

$$\begin{aligned} C &= -N\epsilon \frac{\partial}{\partial T} \tanh\left(\frac{\epsilon}{k_B T}\right) \\ &= \frac{N\epsilon^2}{k_B T^2} \operatorname{sech}^2\left(\frac{\epsilon}{k_B T}\right). \end{aligned}$$

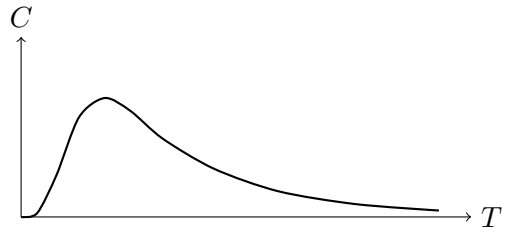
Thus,

$$\boxed{C = \frac{N\epsilon^2}{k_B T^2} \operatorname{sech}^2\left(\frac{\epsilon}{k_B T}\right)}.$$

In the low-temperature limit, $T \to 0$, the particles predominantly occupy the lower energy level, so $C \to 0$.

In the high-temperature limit, $T \to \infty$, the two energy levels become approximately equally populated, and $C \to 0$ again.

Therefore, the heat capacity exhibits a single peak at an intermediate temperature, known as the Schottky anomaly.



2 II

2.1 (4)

With $\epsilon = \mu H$, the internal energy obtained from (2) becomes

$$E = -N\mu H \tanh\left(\frac{\mu H}{k_B T}\right).$$

Meanwhile, the average populations satisfy

$$\begin{aligned} N_\uparrow + N_\downarrow &= N, \\ -\mu H(N_\uparrow - N_\downarrow) &= -N\mu H \tanh\left(\frac{\mu H}{k_B T}\right). \end{aligned}$$

Combining these two equations, we obtain

$$\boxed{N_\uparrow = \frac{N}{2}\left[1 + \tanh\left(\frac{\mu H}{k_B T}\right)\right]}$$

and

$$\boxed{N_\downarrow = \frac{N}{2}\left[1 - \tanh\left(\frac{\mu H}{k_B T}\right)\right]}.$$

Here, $N_\uparrow$ and $N_\downarrow$ represent the average numbers of particles with upward and downward magnetic moments.

2.2 (5)

The magnetization is the average total magnetic moment:

$$M = \mu(N_\uparrow - N_\downarrow).$$

Using the result from (4),

$$\boxed{M = N\mu \tanh\left(\frac{\mu H}{k_B T}\right)}.$$

2.3 (6)

The entropy can be obtained from

$$S = \frac{E - F}{T},$$

where F is the Helmholtz free energy.

From the partition function,

$$\begin{aligned} F &= -k_B T \ln Z \\ &= -N k_B T \ln \left[2 \cosh \left(\frac{\mu H}{k_B T} \right) \right]. \end{aligned}$$

Meanwhile,

$$E = -N\mu H \tanh \left(\frac{\mu H}{k_B T} \right).$$

Therefore,

$$\begin{aligned} S &= \frac{E - F}{T} \\ &= N k_B \ln \left[2 \cosh \left(\frac{\mu H}{k_B T} \right) \right] \\ &\quad - \frac{N\mu H}{T} \tanh \left(\frac{\mu H}{k_B T} \right). \end{aligned}$$

Thus,

$$\boxed{S = N k_B \left[\ln \left(2 \cosh \frac{\mu H}{k_B T} \right) - \frac{\mu H}{k_B T} \tanh \left(\frac{\mu H}{k_B T} \right) \right]}.$$

2.4 (7)

Since the magnetic field is reduced slowly under adiabatic conditions, the process can be treated as reversible and isentropic:

$$\delta S = 0.$$

From (6), the entropy depends only on the dimensionless variable

$$x = \frac{\mu H}{k_B T}.$$

In particular,

$$S = N k_B [\ln(2 \cosh x) - x \tanh x].$$

Differentiating with respect to x,

$$\begin{aligned} \frac{dS}{dx} &= N k_B [\tanh x - \tanh x - x \operatorname{sech}^2 x] \\ &= -N k_B x \operatorname{sech}^2 x. \end{aligned}$$

For $x > 0$, the entropy is a monotonically decreasing function of x.

Therefore, keeping the entropy constant requires

$$x = \frac{\mu H}{k_B T} = \text{constant}.$$

Since μ and k_B are constants,

$$\boxed{\frac{H}{T} = \text{constant}}.$$

Thus,

$$\frac{T_f}{T_i} = \frac{H_f}{H_i}.$$

When the magnetic field is reduced to one-tenth of its initial value,

$$H_f = \frac{H_i}{10},$$

the final temperature becomes

$$\boxed{T_f = \frac{T_i}{10}}.$$

This is the principle of adiabatic demagnetization.

3 III

3.1 (8)

For convenience, let

$$x_i = e^{-\epsilon_i/(k_B T)} \qquad (i = 1, 2).$$

Since each particle can occupy the three energy levels $0, \epsilon_1, \epsilon_2$, the single-particle partition function is

$$Z_1 = 1 + x_1 + x_2.$$

For N distinguishable and non-interacting particles,

$$\boxed{Z = Z_1^N = (1 + x_1 + x_2)^N}.$$

3.2 (9)

The internal energy is

$$E = -\frac{\partial \ln Z}{\partial \beta}.$$

Using

$$\frac{\partial x_i}{\partial \beta} = -\epsilon_i x_i,$$

we obtain

$$\boxed{E = N\frac{\epsilon_1 x_1 + \epsilon_2 x_2}{1 + x_1 + x_2}}.$$

The Helmholtz free energy is

$$\boxed{F = -N k_B T \ln(1 + x_1 + x_2)}.$$

Therefore, using $S = (E - F)/T$,

$$\boxed{S = Nk_B \ln(1 + x_1 + x_2) + \frac{N}{T}\frac{\epsilon_1 x_1 + \epsilon_2 x_2}{1 + x_1 + x_2}.}$$

We now consider the three temperature regions, assuming $\epsilon_2 \gg \epsilon_1$.

(i) $k_B T \ll \epsilon_1$

In this limit,

$$\frac{\epsilon_1}{k_B T} \to \infty, \qquad \frac{\epsilon_2}{k_B T} \to \infty,$$

so

$$x_1 \to 0, \qquad x_2 \to 0.$$

Thus,

$$\boxed{S \to 0}.$$

At sufficiently low temperatures, almost all particles occupy the non-degenerate ground state.

(ii) $\epsilon_1 \ll k_B T \ll \epsilon_2$

In this temperature region,

$$\frac{\epsilon_1}{k_B T} \to 0, \qquad \frac{\epsilon_2}{k_B T} \to \infty.$$

Hence,

$$x_1 \to 1, \qquad x_2 \to 0.$$

The entropy becomes

$$\begin{aligned} S &= Nk_B \ln(1 + x_1) + \frac{N}{T}\frac{\epsilon_1 x_1}{1 + x_1} \\ &\longrightarrow Nk_B \ln 2. \end{aligned}$$

Therefore,

$$\boxed{S \approx Nk_B \ln 2}.$$

In this region, the two lower energy levels are approximately equally populated, while the highest level is effectively unoccupied.

(iii) $k_B T \gg \epsilon_2$

In this limit,

$$\frac{\epsilon_1}{k_B T} \to 0, \qquad \frac{\epsilon_2}{k_B T} \to 0,$$

so

$$x_1 \to 1, \qquad x_2 \to 1.$$

Thus,

$$\begin{aligned} S &= Nk_B \ln(1 + x_1 + x_2) \\ &\quad + \frac{N}{T}\frac{\epsilon_1 x_1 + \epsilon_2 x_2}{1 + x_1 + x_2} \\ &\longrightarrow Nk_B \ln 3. \end{aligned}$$

Therefore,

$$\boxed{S \approx Nk_B \ln 3}.$$

At sufficiently high temperatures, all three energy levels become approximately equally populated.

Combining the results,

$$\boxed{S \approx \begin{cases} 0, & k_B T \ll \epsilon_1, \\ N k_B \ln 2, & \epsilon_1 \ll k_B T \ll \epsilon_2, \\ N k_B \ln 3, & k_B T \gg \epsilon_2. \end{cases}}$$

3.3 (10)

The heat capacity can be calculated from

$$C = \frac{\partial E}{\partial T}.$$

Using the expression for E from (9), we obtain

$$\boxed{C = \frac{N}{k_B T^2} \left[\frac{\epsilon_1^2 x_1 + \epsilon_2^2 x_2}{1 + x_1 + x_2} - \left(\frac{\epsilon_1 x_1 + \epsilon_2 x_2}{1 + x_1 + x_2} \right)^2 \right].}$$

Since $\epsilon_2 \gg \epsilon_1$, the two energy scales are well separated.

At low temperatures, $k_B T \sim \epsilon_1$, the first excited level becomes thermally accessible, producing the first peak in the heat capacity.

At higher temperatures, $k_B T \sim \epsilon_2$, the second excited level also becomes thermally accessible, producing another peak.

In the high-temperature limit, the populations approach equal values, and the heat capacity tends to zero.

Therefore, the heat capacity exhibits two Schottky-type peaks.

